

Ref No.: CECNF/NP/2025-26/138

Roll No. _____

B. Sc. (Hons.) Semester III Examination 2025-26**STATISTICS****STAMV-32 (Practical based on the paper STAMV-31)**

Full Marks: 70

Max. Time: Three Hours

Note: Attempt any 03 question out of the following 04 questions. All the questions carry equal marks.

1. Seven usual and similar Indian coins are tossed and number of tails noted. The experiment is repeated 256 times and the following distribution is obtained:

No. of Tails	0	1	2	3	4	5	6	7
Frequency	14	12	38	70	60	46	14	2

Fit a binomial distribution assuming:

- (i) the nature of the coin is not known.
- (ii) probability of a head for four coins is 0.5 and for the remaining three coins is 0.45.

2. The following table shows the distribution of marks obtained by **500 students** in an examination:

Marks (Less than)	10	20	30	40	50	60	70	80	90	100
Frequency	6	20	56	120	220	340	430	476	494	500

Fit a normal distribution to the above data and obtain the equation of the corresponding normal curve.

3. Out of 20 parcels to be dispatched by a clerk of a post office, eight are to be sent by air mail and the rest by surface mail. The parcels are get mixed up thoroughly and then five parcels are selected randomly. The distribution of the number of parcels marked for air mail getting into the five chosen parcels for a period of 100 consecutive days is reported as follows:

x	0	1	2	3	4	5
f	7	22	45	20	5	1

Calculate expected frequencies by assuming a suitable probability distribution for the above data set.

4. In order to study the number of divorce cases in the western countries, the following distribution is obtained for the time intervals (in years) between the days of their marriage and the day of their divorce:

No. of years	0-5	5-10	10-15	15-20	20-25	25 and above
No. of Persons	190	70	25	10	4	1

Fit an exponential distribution to the above dataset.